



Artificial intelligence approaches for molecular representation in drug response prediction

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Abstract

Drug response prediction is essential for drug development and disease treatment. One key question in predicting drug response is the representation of molecules, which has been greatly advanced by artificial intelligence (AI) techniques in recent years. In this review, we first describe different types of representation methods, pinpointing their key principles and discussing their limitations. Thereafter we discuss potential ways how these methods could be further developed. We expect that this review will provide useful guidance for researchers in the community.

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Introduction

Drug response prediction is essential for the development of personalized medicine. Drug response evaluation in clinical trials is challenging due to the high cost in experiments and time. To overcome the challenge, large-scale datasets of drug response (sensitivity) for cell lines such as Cancer Cell Line Encyclopedia (CCLE), Genomics of Drug Sensitivity in Cancer (GDSC), Cancer Therapeutics Response Portal (CTRP), and Genentech Cell Line Screening Initiative (gCSI), are established [1], enabling the computational prediction of drug response using machine learning and artificial intelligence approaches.

In drug response prediction studies, molecule representation is an important factor that influences model performance. A well-designed representation may bring significant improvement in prediction performance and vice versa. Conventional machine learning methods like linear regression, random forests, and Support Vector Machines (SVMs), have been successfully applied in drug response prediction [2]. However, these methods are based on hand-crafted features/representations and lack the ability of automatically learning molecular representation. In contrast, AI methods especially those recently developed deep learning techniques have the intrinsic ability to learn molecule representation that is relevant to the learning task.

In this work, we focus on the recent development of molecule representation methods. We surveyed related literature published in the last two years. The major types of input data for drugs include sequence, fingerprints (FPs), image, and graph notation. One popular molecular representation is the simplified molecular-input line-entry system (SMILES), which uses a line notation to encode molecular composition and chemical bonds [3]. A SMILES sequence of a molecule usually contains letters, symbols and parentheses, which stands for atoms, bonds and branches, respectively. Molecular FPs include Molecular access system (MACCS) and extended-connectivity fingerprints (ECFPs), both containing molecular structural information. Because drug molecules can be treated as images, image learning methods can also be used to learn molecule representation. Molecular graphs (2D and 3D) are more direct ways to represent molecular structures. In a molecular graph, nodes and edges represent atoms and bonds, respectively. For these different types of representation methods, AI-based solutions can be thought of as feature learning from drug sequences, images, and graphs.

AI methods for molecular representation

We specifically focus on the AI-based molecule representation methods in drug response prediction in the recent two years. Deep learning methods like convolutional neural networks (CNN), recurrent neural network (RNN), graph neural networks (GNN), transformers, contrastive learning, multi-task learning, transfer learning, and pre-trained models have greatly assisted molecular representation and thus improved the

prediction performance of drug response (Table 1). Taking the SubCDR method as an example (Figure 1), it suggests that drug response prediction should consider subcomponents or signatures of drugs and cell lines to improve its accuracy. First, drug subcomponents obtained from the BRICS algorithm and gene subsets are taken as signatures of cell lines. Second, the latent features of drug subcomponents are extracted by gated recurrent unit (GRU), and subfeatures of cell lines are learned by CNN layers. Third, the interaction network of sub-components is constructed by a bilinear scoring strategy. Finally, the embedding representations are obtained from the network through graph convolutional network (GCN), and are used as features with side information to prediction. The root mean square error (RMSE) of SubCDR is 0.9535 and the Pearson correlation coefficient (PCC) is 0.9428 on the GDSC dataset, which outperforms the benchmarking methods such as tCNN [4]. Among the above-described input data types of drugs, molecular FPs represent readily extracted features/representations and the other three types of data, *i.e.* sequence, image and graph, are input data from which AI methods learn molecule representation that are relevant to the task under investigation. So, we divide these methods into three broad categories: (1) sequence-based methods, (2) image-based methods, (3) graph-based methods. In the following sections, we will give a brief introduction to each type of these methods, describing their key principles and discussing their limitations.

(1) Sequence-based methods

For learning numeric features from sequence notations, there are AI-based molecular representation methods proposed in recent years. For example, CNN, RNN and transformers are developed for learning features from drug molecule sequence [5–10]. tCNN [5] first transforms drug sequence into one-hot representation. Then, it utilizes a twin CNN network to learn features from drugs and cancer cell lines, respectively. Finally, with the learned features, a fully connected neural network is applied to predict drug response [5].

Mol2Context-vec uses a deep contextualized Bi-LSTM architecture to learn molecular representation from context awareness for drug discovery [6]. DGSDRP also adopts CNN to capture local chemical context information from drug sequences and extracts the features of cell lines based on CNN and Bi-LSTM [7]. TranGRU integrates RNN and self-attention mechanism into transformer encoders to learn features from SMILES sequence [10]. DeepTTA uses a transformer architecture to learn features of drug substructures and a four-layer neural network to extract features of cell lines from gene expression data [9]. iBT-Net is the first work to unite transformer encoder and broad learning framework in cancer drug response (CDR) prediction and achieves better performance (Table 1) [11]. TranGRU, DeepTTA,

IBT-Net and SubCDR share some commons with considering subcomponents of drug into transformer encoder or GRU to learn latent features of drugs (Figure 1). Of note, a recent study exploits the emerging large language models for molecule representation [8]. The authors propose MOLFORMER which utilizes transformer-based language models to train SMILES sequences of 1.1 billion unlabelled molecules to learn spatial relationships between atoms. The learned representation can be further used for drug response prediction.

Although sequence format can be processed by using natural language processing (NLP) methods, molecule sequence is much more complicated than conventional NLP problems. Using only SMILES sequence is hard to capture the spatial relationship between atoms and functional groups, so the current methods usually supplement other types of molecular information. In the future, methods based on molecular sequence may need to pay more attention to the representation of the spatial or positional information, and the relationship between atoms and substructures in the drugs.

(2) Image-based methods

Different from the previous methods that treat SMILES as text inputs, some studies treat the input as images. Then, drug response prediction is formulated into an image classification or prediction task. For example, REFINED-CNN considers the 2D image of drugs as input to predict drug sensitivity by stacking CNNs [12]. ImageMol is proposed to apply unsupervised image-based pretraining models for computational drug discovery [13]. In this method, the images of 10 million unlabelled drug-like, bioactive molecules were first pretrained to obtain the chemical representation of molecules. The learned representation achieves high performance in multiple specific tasks including the prediction of drug targets and drug metabolism.

However, image-based methods are confronted with several issues. First, the molecular image does not contain the additional chemical attributes of atoms and bonds. Second, molecules have different sizes and this may cause scale problem in image learning. As mentioned above, likely due to these issues, image-based methods are not main molecular representation methods in the present.

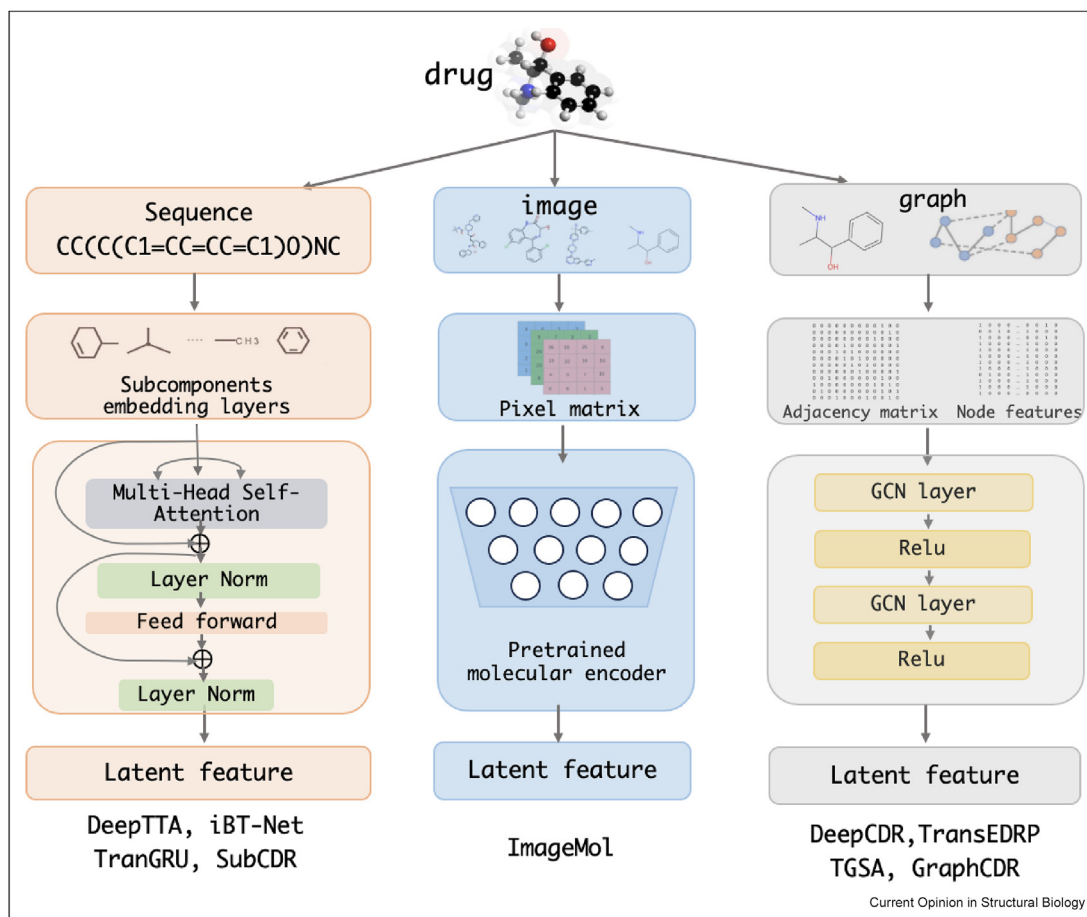
(3) Graph-based methods

Graph-based methods consider a molecule as a graph, which encodes structure information more directly. These methods include GCN, graph isomorphism network (GIN), and graph attention networks (GAT). Many researchers seek to capture more accurate molecular representation by GNN. Usually, a molecular graph is presented by an adjacent matrix and a feature

Table 1**The benefits, limitations, and performance of molecular representation methods.**

Type of method	Study	Dataset	Molecular representation approach	Metrics	Benefits	Limitations
Sequence-based methods	tCNN [5]	GDSC	CNN (Drug in canonical SMILES is converted to 72*188 one-hot matrix)	RMSE: 1.228 PCC: 0.909	tCNN performs better when IC50 value is around the median.	tCNN overestimates when IC50 value is small and underestimates when IC50 value is big. Inexplicability and not consider multi-omics features.
	DeepTTA [9]	GDSC	Transformer with multi-attention layers (the molecular sequence is segmented into a substructural sequence by ESPF)	RMSE: 0.952 PCC: 0.941	DeepTTA considers substructure and is the first work to utilize transformer architecture.	
	iBT-Net [11]	GDSC2	The molecular feature is comprised of a transformer encoder block and a MLP block (a molecular sequence is preprocessed into a length of 183 substructures)	RMSE: 0.924 PCC: 0.942	iBT-Net is the first work to unite deep learning and broad learning framework in CDR prediction and achieves better performance.	
	MOLFORMER [8]	MoleculNet, PubChem, ZINC	Transformer with linear attention relative position Embeddings (MOLFORMER takes molecular sequences as inputs.)	AUC: 0.937 (BBBP)	MOLFORMER includes positional embedding and can be used for molecular representation for agnostic task.	The misuse of MOLFORMER without web lab validation might lead to misleading results.
Image-based methods	ImageMol [13]	MoleculNet, PubChem	ResNet18 (Molecular images are transformed and set into size of 224 × 224 by RDKit.)	AUC: 0.952 (BBBP)	ImageMol uses pretrained models to capture molecular representative feature.	ImageMol cannot capture 3D information.
Graph-based methods	GraphCDR [17]	GDSC, CCLE	GCN with contrastive learning	RMSE: 1.0123 PCC: 0.9341	GraphCDR utilizes contrastive learning task as a regularizer to boost CDR prediction.	Not consider diverse biological association information.
	TGSA [16]	GDSC2, CCLE and COSMIC	GCN with pre-training strategy (ZINC15 dataset is used in the pre-training stage to transfer the domain knowledge of chemistry into node level feature)	RMSE: 0.877 PCC: 0.951	TGSA considers prior domain knowledge of PPI and treats CDR as edge prediction problem.	Not consider node features.

Figure 1



Molecular representation methods based on sequence, image, and graph.

matrix (each node has a feature vector), which are the two required inputs of GNN. In recent years, GNN-based models are shown to be effective in drug response prediction.

In GraphDRP, four variants of GNN including GCN, GIN, GAT and GCN_GAT are tested for drug representation. It is found that the results of graph-based methods performs better than tCNN, a representative sequence-based method [14]. For example, on the prediction for unseen cell-line experiment using the GDSC dataset, GraphDRP achieves significantly lower RMSE (0.0358) than that (0.0576) of tCNN. DeepCDR proposes a uniform GCN to learn latent features of drug after extracting molecular FPs of atoms through “deepchem” and Rdkit [15]. TGSA uses two GNN encoders, i.e. GIN and GAT, to learn the representation of drugs and cell lines, respectively [16]. In TGSA, pre-training strategies are applied among the procedure of GIN and similarity augmentations are proposed to

enrich the embedding of drug and cell line through GraphSAGE [16]. In GraphCDR, deep neural network and GCN are employed to learn node representation, and contrastive learning strategies are then adopted to enhance the feature representations of drugs and cell lines [17].

Besides, graph transformer networks (GTN) are also introduced to drug response prediction. GTN is a generalization of transformer networks for arbitrary structures of graphs [18], which is capable of generating new graphs and might capture more enriched information from graph. GraTransDRP [19] is proposed to use GTN to overcome the limitation of graph-based methods in drug feature learning from large-scale multi-omics data. TransEDRP adopts a dual transformer model to predict drug response. In TransEDRP, drug’s representation is obtained by learning both edge features and node features through a graph transformer module.

In addition to features of drugs, features of other omics data can also be learned and integrated for drug response prediction. For example, MOFGCN predicts drug response by using multi-omics fusion [20]. More specifically, MOFGCN calculates cell line similarity, drug similarity and cell line-drug association based on multi-omics data. Then, GCN is performed on a heterogeneous network, which is built by integrating the above three networks, to learn features. Finally, a correlation matrix between drugs and cell lines is constructed to predict drug response [20].

Future perspective

The past two years has seen substantial advancement of methods in drug response prediction. For the future studies, AI methods may incorporate information relevant to drug efficacy to enhance existing methods. In this brief review, we are not aimed to provide a comprehensive overview of how the methods in this field can be improved, but only focus on the three aspects that we think are of interest, namely 3D spatial structure integration, knowledge graph, and multimodal data integration, as described below.

First, because 3D structures determine drug responses, integrating 3D spatial structure offers a promising way for drug response prediction. Taking graph-based methods as an example, they are flexible and can be developed further by integrating 3D spatial information of drugs that may be predictive of response. For example, GeoGNN [21] is geometry-based graph neural network architecture and is pretrained to learn the geometries from molecules with 3D spatial structures.

Second, knowledge graph has become popular in recent years for its ability to modeling relationship between entities, and it has been shown to facilitate the study of drug response and drug discovery [22–26]. MoCL utilizes global-domain and local-domain knowledge to improve molecule representation through contrastive learning on molecular graphs [23]. KANO first integrates knowledge of elements and functional groups. Then, it uses a contrastive learning framework to capture the graph representation of molecules based on the knowledge [25]. Otter-Knowledge incorporates multiple knowledge graphs and trains multimodal GNN to enrich SMILES representations [26]. In summary, the knowledge about molecules usually includes sequence information, structural information, chemical properties of elements and functional groups, spatial information of molecules from pretrained models or public databases, and similarity networks of drug-drug or drug–target.

Third, as the data in different modals may complement each other in improving drug response prediction, multimodal data integration has become an essential

direction in the community [27–32]. In addition to learning contextual features from sequences of drugs, DGSDRP can also consider topological structure information from molecular graph data by using GCN [7]. Liu *et al.* proposes a unified 2D and 3D pre-training model to represent features of molecules [27]. Using this molecular representation method, the authors perform the downstream task of molecular property prediction. The results show that the method performs better in most cases [27]. MM-Net uses data augmentation and multimodal neural networks to predict drug response [28]. Modality-DTA [29] is a novel multi-modality fusion strategy that considers six modalities of drug features and target features. DeepCoVDR [30] is a deep learning framework for predicting COVID-19 drug response based on three methods including transfer learning, GTNs and cross-attention. RAMP utilizes response-aware negative sampling to improve network embedding, and then represents a multi-task learning framework of Bayesian neural networks with contrastive regularization to predict drug response. This method can overcome the imbalance problem compared to the other methods [31].

With the development of AI techniques, methods in the future are not limited to those mentioned above. Graph-based methods with multi-task learning, GTN, pre-trained models, contrastive learning, and transfer learning, may be promising in drug response prediction.

Conclusion

In the recent two years, AI approaches have greatly promoted the development of molecule representation in drug response prediction. Despite the achieved success, challenges still exist for predicting drug response. For example, AI methods for 3D molecular representation with geometry information, which is essential for understanding drug activity, are still in demand. Besides, current AI-based models are still limited at interpreting key causal information about drug response and the binding mechanisms between drugs and targets such as cell lines, tumors, chemicals, etc. We expect that the development of novel AI methods will boost molecular representation and thus drive forward the area of drug response prediction.

Declaration of competing interest

None declared.

Data availability

No data was used for the research described in the article.

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